Mansaa Kohli

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EDUCATION

Vellore Institute of Technology Sep 2023 – May 2027
B.Tech in Artificial Intelligence and Machine Learning CGPA: 9.07 / 10.0

TECHNICAL SKILLS

Languages: Python, Java, SQL
Core CS: Data Structures and Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems
Problem Solving: 250+ LeetCode problems solved; Codeforces contest participant
Backend and Databases: FastAPI, REST APIs, JWT Authentication, PostgreSQL, SQLAlchemy, Supabase
Frontend: React, HTML, CSS
Machine Learning and GenAI: Machine Learning, Scikit-learn, XGBoost, LightGBM, LLM APIs, LangChain, Retrieval-Augmented Generation (RAG), Vector Databases (ChromaDB), Embeddings, Prompt Engineering
Data Analysis: Pandas, NumPy, Matplotlib, Data Visualization, Data Preprocessing, Model Evaluation, Cross-Validation, Hyperparameter Tuning
Cloud and Tools: AWS, Git, GitHub, Pytest, Vercel, Render, Railway

PROJECTS

RepoIQ – GitHub Repository Analysis Platform
FastAPI, React, TypeScript, PostgreSQL, Groq API (Llama 3.3 70B) repoiq-insight.vercel.app

- Built a full-stack web application that analyzes any public GitHub repository and uses a Large Language Model (LLM) to generate architecture summaries, tech-stack breakdowns, README drafts and interview questions.
- Designed a transparent rule-based health score (0–100) from 10 weighted signals, including tests, CI workflows, commit recency, license and documentation, and presented results in an analytics dashboard.
- Architected a layered FastAPI REST backend in Python (routers, services, repositories) with JWT/bcrypt authentication, SQLAlchemy 2.0 and PostgreSQL; wrote Pytest unit tests and deployed on Vercel and Render.

Launchpad – AI Career Preparation Platform
Next.js, FastAPI, LangChain, ChromaDB, Groq API (Llama 3.3 70B) launchpad-smoky-ten.vercel.app

- Built an ATS resume analyzer that scores resumes (0–100) against a job description, flags missing keywords and exports ATS-formatted PDFs.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline (LangChain chunking, MiniLM embeddings, ChromaDB vector search) for source-grounded answers from course PDFs.
- Developed a 5-round adaptive mock interview with per-answer scoring (0–10), feedback and voice input; deployed on Vercel and Railway.

MafiaMind – Real-Time Multiplayer Social Deduction Game
React 19, TanStack Start, TypeScript, Supabase, Google Gemini mafia-mind.vercel.app

- Built a real-time game (up to 8 players) on Supabase Realtime where Gemini-powered AI players chat and vote.
- Engineered structured LLM outputs (0–100 suspicion scores with reasoning) that drive AI voting and a heatmap.
- Secured hidden roles with Postgres row-level security, server-authoritative logic and role-reveal database triggers.

EXPERIENCE

EDXcellence Jan 2026 – Apr 2026
AI with Python Intern Remote

- Fixed a class-precision gap (0.61 vs 0.79) in a diabetes Random Forest model by handling zero-encoded missing values and class imbalance in a scikit-learn pipeline; recall rose 0.62 to 0.76, ROC-AUC 0.812 to 0.837.
- Benchmarked 9 regression models using 5-fold cross-validation and GridSearchCV hyperparameter tuning on 1,000 records; tuned Ridge reached R-squared 0.88 (RMSE 5.37); shipped a CLI for single and batch predictions.

ACHIEVEMENTS, PUBLICATION AND CERTIFICATIONS

- **3rd Place, CodeVerse Hackathon 2025** (200+ teams): built SafeWalk, an in-browser AI fall-detection app.
- **Selected for Amazon ML School 2026** (top 3,000 of 134,000+ applicants).
- **Co-author, ICAAIC 2025:** LightGBM vs deep learning for diabetes onset prediction (96.8% accuracy).
- **Certifications:** Azure Data Fundamentals (DP-900); IBM Generative AI (watsonx); UMich Applied ML in Python.